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Working Group X - Outflows

Welcome to Group X of Project AGORA! This group will focus on the science of gas flows in galaxies, in particular galactic outflows and feedback processes, the circumgalactic medium (CGM), and the intergalactic medium (IGM) in high resolution cosmological simulations.

Science questions

This is a partial list of questions compiled by Michele Fumagalli and Sijing Shen. Please add your thoughts!

a. The circumgalactic medium:

- How does the CGM get enriched? How far does the enrichment extend to? What are the spatial distribution, thermodynamics and kinematics of enriched gas?
- How well do simulations compare to observations of HI and metal absorption systems (e.g. Si II, CIV, MgII and OVI etc.)?
- How dose the CGM and its metals evolve with time and vary with galaxy mass and star formation history?

b. The outflows:

- What are the properties of outflows, such as metallicity, mass loading, velocity and temperature? What is the percentage of outflowing gas re-accreted back?
- How do outflow properties depend on galaxy mass, SFR, feedback mechanisms and gas cooling?
- Are outflows degenerate with respect to other physical mechanisms of galaxy formation? In other words, what is it the 'minimal' prescription for outflows that does not violate direct observational constraints?
- How does outflowing gas interact with the inflowing material? Can inflows be observed with the presence of outflows?

c. The impact on galaxy formation and evolution:

- What are the roles of galactic outflows on the regulation of star formation, on galaxy morphology and other galaxy properties?
- If feedback properties vary with galaxy mass and SFR, can we learn something from zoom-in simulations about the luminosity function?
- What types of galaxies are more likely affected by outflows? What is the relative importance of galactic outflow comparing to other feedback mechanism such as photoionization heating from stars?

d. Variation between different participating codes:

- Using the same (or similar) physics, do different codes yield different results (especially between Eulerian and Lagrangian methods)? For example, the spatial distribution and covering factor of metal absorbers may depend on how well metals are mixed into the CGM, which may vary from code to code.

Leadership

Sijing Shen wish to lead the collaboration of this team, please speak up if you are also interested.

Code representation

Below is an incomplete list of code representation in our group. **Please sign up or correct it.**

Codes	Cosmological Simulators	Code Contacts in Group X
Arepo	?	Dusan Keres
Art-NMSU	Daniel Ceverino	Daniel Ceverino, Michele Fumagalli
Art-Chicago	Sam Leitner	Sam Leitner
Enzo	John Wise, Cameron Hummels	Cameron Hummels, Britton Smith, Greg Bryan
Gadget	Amit Kashi, Dusan Keres, Justin Read	Amit Kashi, Dusan Keres, Phil Hopkins, Ken Nagamine
Gadget-SPHS	Justin Read	?
Gasoline	Sijing Shen	Javiera Guedes, Piero Madau, Sijing Shen, James

		Wadsley
Ramses	Oscar Agertz, Romain Teyssier	Oscar Agertz, Romain Teyssier
Nyx	Wolfram Schmidt	?

As of now, we don't have members using Gadget-SPHS and Nyx. People within the Agora Project who use these codes will be contacted.

Working Group X - Blog

Untitled Post This is the first blog post for the Working Group X – Outflows.

Posted Oct 4, 2012, 12:53 AM by Ji-hoon Kim

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